

The Logarithmic Constraint: Modeling Quantum Avoidance as a Non-Linear, Context-Sensitive Morphism

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Abstract

The Pauli Exclusion Principle (PEP) is traditionally interpreted as a physical constraint preventing identical fermions from occupying the same quantum state. We propose a novel computational framework that reinterprets this principle as a logical axiom of “inverse tiling,” where the goal is to enforce absolute non-identity between states. We present the **Avoidance Wave Morphism (AWM)**, a highly efficient algorithm that generates probability distributions with unique spectral signatures. By utilizing a context-sensitive, non-linear indexing formula applied to a random walk, the algorithm scales with logarithmic complexity relative to output resolution while maintaining a constant Signal-to-Noise Ratio (SNR). Spectral analysis of over 10^{11} samples reveals a complete absence of harmonics, confirming that the algorithm successfully enforces spectral uniqueness and non-degeneracy, offering a computational existence proof for quantum avoidance behavior.

1 Introduction

In quantum mechanics, the Pauli Exclusion Principle (PEP) dictates that two identical fermions cannot occupy the same quantum state simultaneously. While often described via the antisymmetry of the wavefunction, the underlying logical structure can be viewed as an axiom of **uniqueness**.

If we consider a phase space filled with potential distributions, the PEP acts as the “inverse of collision”: it is a constraint that ensures no two distributions (or “tiles”) are identical in phase and amplitude. This paper introduces a computational model, the Avoidance Wave Morphism (AWM), which generates such unique distributions. Unlike classical linear systems, AWM relies on a context-sensitive transformation that is computationally economical, scaling logarithmically, yet producing spectrally rich, non-degenerate outputs.

2 The Avoidance Wave Morphism (AWM)

The core of the model is a specific algorithm that maps a cumulative random walk to a final probability distribution index. The transformation is non-linear and context-sensitive, meaning the processing rules change based on the target resolution.

2.1 Context-Sensitive Parameters

Let N be the initial number of bins (resolution). The algorithm defines two scaling parameters that serve as the “context” for the morphism:

1. **Effective Steps (n_{bins}):** The length of the random walk scales logarithmically with resolution, compressed by the natural log and the square root of π .

$$n_{\text{bins}} = \lfloor \ln(N \cdot 6) \sqrt{\pi} \rfloor \quad (1)$$

2. **Magnitude Factor (RN_{mag}):** A non-linear scaling factor for the random sum.

$$RN_{\text{mag}} = \left\lfloor \frac{N}{\sqrt{\log_2(N)}} \right\rfloor \quad (2)$$

2.2 The Indexing Morphism

The transition from the random walk to the final state is governed by a non-invertible morphism $\mathcal{M}$. Let S be the sum of n_{bins} random numbers generated by a chaotic PRNG (specifically *phimagic_prng32*). The final index I is computed as:

$$I = \mathcal{M}(S, N) = ((S \times RN_{\text{mag}}) \gg 32) \pmod{N} \quad (3)$$

where $\gg 32$ denotes a right bit-shift of 32 bits. This operation acts as a chaotic filter, discarding lower-order bits and mixing the remaining information based on the context RN_{mag} .

2.3 Algorithm Implementation

The Python implementation of the core logic is as follows:

```

1 # Context-Sensitive Scaling
2 n_bins = np.uint64(np.round(np.log(initial_n_bins * 6) * np.sqrt(np.pi)))
3 rn_mag = np.uint64(np.round(initial_n_bins / np.sqrt(np.log2(initial_n_bins
4 )))
5 # The Morphism Loop
6 for _ in range(trials):
7     random_walk = np.uint64(0)
8
9     # Accumulate Random Phase
10    for _ in range(n_bins):
11        random_walk += np.uint32(prng.rng())
12
13    # Non-linear Indexing (The Avoidance Filter)
14    index = np.uint32(np.mod(random_walk * rn_mag >> np.uint32(32),
15        initial_n_bins))
16
17    cycle[index] += 1

```

Listing 1: The AWM Core Logic

3 Results: Spectral Uniqueness

To verify the hypothesis that this morphism enforces the “inverse of tiling” (non-identity), a Fast Fourier Transform (FFT) was performed on the generated distributions.

3.1 Absence of Harmonics

Analysis of the output generated from 10^{11} samples demonstrated a significant spectral property: **No detectable harmonics were found in the frequency domain.**

In classical signal processing, harmonics imply periodicity and redundancy. The absence of harmonics suggests that the generated probability distribution is:

- Non-periodic.
- Composed of incommensurate frequencies.
- Mathematically unique for every distinct set of initial parameters.

This spectral signature confirms that the AWM successfully creates “Avoidance Waves”—states that are maximally distinct from one another, satisfying the uniqueness constraint of the Pauli Exclusion Principle.

4 Discussion

4.1 Logarithmic Efficiency and Information Compression

A critical observation of the AWM is the relationship between computational cost and resolution. As observed in Eq. (1), the number of random steps grows logarithmically:

$$\text{Complexity} \propto \frac{\ln(\text{Resolution})}{\text{Resolution}} \quad (4)$$

despite this reduction in relative steps, the Signal-to-Noise Ratio (SNR) of the resulting distribution remains stable. This implies that the algorithm achieves a high degree of **information compression**. The logic of avoidance is not enforced by brute-force redundancy (as in a linear system), but by the efficient, context-sensitive mixing of the bit-shift morphism.

4.2 The Run-Time Morphism

We argue that the AWM represents a class of mathematical objects that can be *run* but not easily *proven* via closed-form analysis. The interaction between the chaotic PRNG and the non-linear bit-shift creates an output that is deterministic yet analytically intractable. This mirrors the nature of Turing-complete systems where the final state is known only through execution. In this framework, the Pauli Exclusion Principle is not just a static law, but an active, executable algorithm of nature.

5 Conclusion

We have presented the Avoidance Wave Morphism, a statistical model that enforces the logical axiom of quantum non-identity. By demonstrating that a context-sensitive, logarithmically scalable algorithm can produce unique, non-harmonic probability distributions, we offer a new perspective on fermionic avoidance. This suggests that the fundamental “inverse of collision” may be rooted in principles of extreme computational economy and unique spectral tiling.

6 Mathematical Formulation

To rigorously describe the dynamics of the AWM, we formulate the algorithm as a solution to a *Logarithmic Diffusion Equation with Non-Linear Projection*.

Let $u(\xi, \tau)$ denote the probability density of the random walker in an unbounded space ξ over a logarithmic time scale τ . The system is governed by the diffusion equation:

$$\frac{\partial u}{\partial \tau} = \mathcal{D} \nabla^2 u, \quad \text{subject to } 0 \leq \tau \leq \tau_{\max} \quad (5)$$

where $\tau_{\max} = \ln(N \cdot 6) \sqrt{\pi}$. The final probability distribution $\rho(x)$ is obtained by applying the **Non-Linear Projection Operator** $\hat{\mathcal{M}}$, which corresponds to the bit-shift and modulo operations in the algorithm.

The complete **Avoidance Wave Equation** is thus given by:

$$\rho(x) = \int_{\mathbb{R}} \mathcal{K}_{\text{fold}}(x, \xi; N) \cdot \left(e^{\tau_{\max} \nabla^2} \delta(\xi) \right) d\xi \quad (6)$$

Where:

- $e^{\tau_{\max} \nabla^2}$ is the **Time-Evolution Operator** for logarithmic diffusion, generating the raw entropic potential.
- $\mathcal{K}_{\text{fold}}$ is the **Avoidance Kernel** representing the context-sensitive morphism (the chaotic bit-shift map) that projects the diffused state onto the finite domain $[0, N]$ while enforcing spectral uniqueness.